

SMARTBRIDGE

SKILL WALL

No. of Functional Features Included in the Solution

Date: 30 September 2026
Team ID: SWTID-2026-6777
Project Name: ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks: 5 Marks

Functional Features Overview:

This document lists the functional features included in the ComicCraft project solution. Each feature describes its purpose, implementation module/component, status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Story Generation	Creates comic story content from a user's story prompt using Gemini-generated narrative output.	gemini_flash.py / story generation route	Done	0.75
2	Character & Scene Creation	Transforms story ideas into structured characters, scenes, dialogue, and visual descriptions for comic panels.	gemini_pro.py / schemas.py	Done	0.75
3	Comic Panel Generation	Builds individual comic panels from generated story scenes and visual prompts.	image_generator.py / panel workflow	Done	0.75
4	Comic Layout Builder	Arranges generated panels, captions, dialogue, and scene elements into a readable comic layout.	layout_builder.py	Done	0.50
5	Comic Preview	Displays the generated comic in a web interface so users can review the story and panels before export.	templates/comic_preview.html + static assets	Done	0.75
6	PDF Export	Exports the generated comic as a PDF document for saving and sharing.	exporters.py	Done	0.50
7	Gemini API Integration	Connects the application to Google Gemini models through centralized configuration and model modules.	gemini_flash.py / gemini_pro.py / config.py	Done	0.50
8	Web Application Interface	Provides story input, generation controls, preview, error handling, and export flows through the FastAPI web application.	app/main.py + templates + static/css + static/js	Done	0.50

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features

Responsive visual styling, animated UI effects, improved comic themes, additional export formats, richer panel layouts, and future multilingual/customization support.

Features Tested & Verified

The core application flow was checked for story prompt submission, Gemini-based generation, comic panel creation, layout building, preview, PDF export, API integration, and error handling.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Story input, generation controls, comic preview, and export interface.
2	Backend / Logic	4	Story generation, character/scene structuring, panel generation, and layout building.
3	File / Export	1	Generated comic PDF export and application export handling.
4	API / Integration	1	Google Gemini API through centralized model/configuration modules and FastAPI routes.
5	Image Generation	1	Comic panel image generation through image_generator.py.
6	Security / Configuration	1	GEMINI_API_KEY is loaded through environment configuration; secrets are not hard-coded in application source.

ComicCraft Solution Focus

ComicCraft is an AI-powered comic story creator using Gemini models. The solution accepts a story idea, generates structured comic content, creates visual panel content, assembles the panels into a comic layout, provides a preview, and supports PDF export.

Project Identification

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